

Qihao Shao

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Research Interests

- LLM agents, Agent Memory, and tool-use agents.
- Uncertainty estimation, model calibration, and reliable decision evaluation.
- Benchmark design, long-horizon task evaluation, and engineering evaluation systems.

Education

Dalian No. 24 High School	2021.09 – 2024.06
High School	
University of Chinese Academy of Sciences, Cybersecurity	2024.09 – Present
Undergraduate	

Research Experience

Agent Memory Benchmark	2026.06 – Present
Causal Ripple Effect in Agent Memory Research in Progress	
• Surveyed HorizonBench, STALE, PersonaMem-v2, and LongMemEval, focusing on task formulation, data construction, and evaluation blind spots. • Summarized data construction patterns for long-term memory evaluation, including state-first generation, old/new observation pairs, and pre-evolution hard negatives. • Contributed to an Agent Memory evaluation framework covering preference evolution, stale-memory suppression, over-personalization, and premise resistance. • Organized evaluation signals such as evolved-static gap, old-preference distractor rate, and scenario applicability judgment.	

Projects

Qwen-vLLM Inference Service Optimization	2026.06 – 2026.07
Pilot Cup, National College Student Computer System Capability Competition Team Lead	
• Built a Qwen inference optimization branch on top of vLLM under competition constraints, focusing on long-context prefill, decode, KV cache behavior, sampling hot paths, and custom operator overhead. • Used Hippof and runtime artifacts to analyze attention paths, TTFT, TPOT, and throughput bottlenecks. • Improved official benchmark throughput by 143% for 16–32K, 75% for 8–16K, and 49% for 4–8K; final rank: 57/346.	

Skills

Programming	Python, C/C++, JavaScript
Deep Learning	PyTorch, Transformers, Hugging Face, vLLM
Agent / LLM	Prompt Engineering, Tool-use Agent, RAG, Memory Mechanism, LangChain
Evaluation	Benchmark Design, Metric Design, Ablation Study, Error Analysis, Calibration Analysis
Engineering	Git, Linux, Docker, Conda, LaTeX
Data	NumPy, Pandas, Matplotlib, JSON, SQLite

Honors

- Outstanding Communist Youth League Member, University of Chinese Academy of Sciences, 2025.
- Third-class Undergraduate Scholarship, University of Chinese Academy of Sciences, 2025.